

Safe Band: IoT-Based Smart Wearable for Emergency Health Alerts and Monitoring in Sri Lanka

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Project Proposal Report

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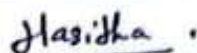
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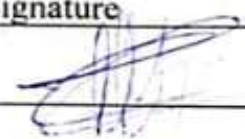
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Cardiovascular disease (CVD) constitutes one of the major causes of mortality in Sri Lanka, with heart attacks usually occurring in people who have several silent risk factors such as high blood pressure, diabetes, tobacco use, genetic predisposition, and obesity without prior systematic evaluation before deploying wearables. Current wearables like smartwatches only respond to vital sign irregularities in real time, failing to consider preemptive prevention classification from baseline patient health profiles.

This paper introduces the Predictive Component of the Safe Band IoT wearable initiative: a pre-deployment heart attack risk classifier. Prior to the patient receiving the Safe Band IoT wearable device, they undertake a Google Form questionnaire with demographics (age, gender, height, weight), lifestyle (smoker, exercise frequency), medical (heart disease history, diabetes, hypertension, medication), and self-rated health data. These data are transformed into features and fed to a Decision Tree Classifier algorithm constructed using the scikit-learn package in Python. The model generates a YES/NO binary risk classification outcome either High or Low Risk which determines the degree of monitoring by the wearable.

The decision tree algorithm takes into consideration clinically verified risk factors such as hereditary heart disease, age more than 50 years old, smoker, hypertension, diabetes, high cholesterol levels, high BMI, and inactive lifestyles. The machine learning model is trained on the UCI Heart Disease Dataset through 10-fold cross-validation. The model is embedded with the Flask REST API framework for integration into the project's MERN stack backend architecture.

The expected results include early identification of risks in patients in resourceconstrained settings in Sri Lanka, lower rates of false negatives compared to traditional clinical assessments, and adaptable wearable device behaviors based on predicted risk categories. This component contributes to achieving the UN Sustainable Development Goal 3 (Good Health and Well-being) through preventive care at scale.

Keywords: Decision Tree, Heart Attack Risk Prediction, IoT Wearable, Safe Band, Cardiovascular Disease, Sri Lanka, Machine Learning, Pre-deployment Screening.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
AUC	Area Under the Curve
BMI	Body Mass Index
CVD	Cardiovascular Disease
ECG	Electrocardiogram
IoT	Internet of Things

ML	Machine Learning
MERN	MongoDB, Express.js, React, Node.js
SDG	Sustainable Development Goal
SpO2	Peripheral Capillary Oxygen Saturation
UCI	University of California Irvine
WHO	World Health Organization

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1. INTRODUCTION

1.1 Background and Context

The Real World Problem and Impacted Stakeholders: In the context of Sri Lanka, CVDs are one of the top causes of sudden death; people affected are usually unaware of their risk

factors. While the technology used in a "Safe Band" helps monitor a patient in real-time, there still is an issue that can be called an "assessment gap," where devices are used without an understanding of the particular user's risk profile. This means that currently patients use the "one size fits all" monitoring approach, meaning that both the person with a high risk (a smoker with hereditary predisposition, etc.) and a low-risk individual are monitored in a similar way. The issue mainly targets the patients living in conditions where they cannot undergo a thorough examination as well as healthcare workers responsible for allocating resources effectively.

Statistical Evidence: As stated by WHO, more than of all deaths in Sri Lanka are associated with CVDs. The study also shows that using risk stratification before the cardiac event happens and giving high-risk individuals increased monitoring (for example, increasing the frequency of physiological sampling) improves survival rates of the patient. Using indicators like BMI, smoking, hereditary predisposition, etc., as predictors can help accurately identify at-risk candidates.

Current Solutions and Their Limitations: Currently, patients in Sri Lanka use either clinical visits or sophisticated smartwatches for health screening. Both methods suffer from three main issues:

High entry barrier: blood tests and clinical visits require expensive equipment, as well as lots of time which is not always available to the patients in rural areas.

Reactive IoT design: current designs usually provide notifications only after a heart rate spike, ignoring the patient's behavior which could increase this risk.

Lack of integration: static information (weight, smoking habits, hereditary factors) is not taken into account while using the device itself.

Proposed Solution: Heart Attack Risk Prediction Component: To overcome this problem, the presented solution includes the creation of a Pre-Deployment Risk Classifier. Prior to providing users with a device, the patient fills out a digital intake form containing such variables as Name, BMI (which is derived from Height/Weight), Smoking, Hereditary risk, and personal health condition. Using Decision Tree algorithm for machine learning, the patient is classified as "high-risk" or "low-risk." This classification will help customize the monitoring process for the individual patient's needs.

1.2 Literature Review

Analysis of Existing Research Approaches: Numerous scientific papers have been published on using the UCI Heart Disease dataset to train models [1]. For example, Mohan et al.'s research shows that while Deep Learning allows obtaining high accuracy, the Decision Tree algorithm is more appropriate for healthcare purposes since it is "interpretable" medical professionals understand why a person was identified as having high risk, for instance, "Age

> 50 AND Smoker = Yes." Nevertheless, most studies remain at the "offline" stage only a few scientists explore the application of these predictions for dynamic IoT hardware configuration in runtime.

Existing Products and Services: Some commercial solutions such as the Apple Watch Series 9 and Fitbit Sense support highly sophisticated ECG and SpO2 measurements [2].

Unfortunately, these devices represent the "Lifestyle category" and do not inquire about a customer's hereditary background and tobacco use to forecast any future events. They only provide real-time information about their physical state. Also, a few mobile applications implement "Heart Risk Calculators." However, those applications operate independently and cannot be combined into a cohesive wearable monitoring ecosystem like Safe Band.

Limitations of Existing Systems: 1. Inability to Explain: Sophisticated algorithms are like "Black Boxes," which makes healthcare specialists skeptical about their predictions [3].

2. Isolated Ecosystems: The "Risk Prediction" system and the "Real-Time Monitoring" system run separately, and there is no exchange of data between the "Intake Form" and the "Wearable Alert Logic."

3. Inefficient Hardware Utilization: Wearable devices sample healthy people as often as vulnerable subjects, leading to unnecessary power drain.

1.3 Research Gap

The Missing Element in Clinical Surveillance: The first missing element of Sri Lankan Healthcare Technology infrastructure is a preemptive stratification layer of the IoT wearables system. There is no technology that captures basic noninvasive lifestyle information (through a questionnaire form) to "prime" a smart wearable to perform customized emergency detection monitoring.

Novelty and Creativity of the Proposed System: The uniqueness of this particular innovation stems from its Intake to Hardware integration. In contrast to the available approaches, this system provides the following novel components:

Non-Invasive Predictive Analysis: Decision Tree algorithm for providing a highly accurate risk probability score without having any blood sample or clinical visits at all.

Dynamic Configuration: Prediction Algorithm results directly dictate changes in Smart Band's code execution (for example, High Risk one minute interval sampling, and Low Risk five minutes interval sampling).

Localized Feature Engineering: Assigning weightages to features such as "Hereditary History" and "Smoking." Both are extremely prevalent risk factors among the local demographics.

Evidence-Based Argument: Scientific evidence demonstrates that people who have a hereditary history of CVD conditions have higher chance of experiencing a myocardial infarction. With the help of a Google Form and an ML model, the Safe Band becomes not just a reactive monitoring tool, but a proactive, personalized, and preventive healthcare system.

2. OBJECTIVE

2.1 Main Objective

To develop and deploy a Decision Tree-based machine learning classifier that predicts individual heart attack risk from a structured patient questionnaire, enabling automated predeployment risk stratification and wearable device configuration within the Safe Band IoT health monitoring system.

2.2 Specific Objectives

1. To design and implement a structured Google Form data collection instrument capturing eight cardiovascular risk factors (hereditary heart disease, age, smoking status, BMI, diabetes, hypertension, cholesterol level, and physical activity) and map these into ML-ready binary and ordinal features.
2. To build and train a Decision Tree Classifier using Python's scikit-learn library on the UCI Heart Disease Dataset, achieving a validation and an AUC-ROC via 10 fold cross validation.
3. To implement a Flask REST API that accepts patient form submissions as JSON, processes features, runs the trained model, and returns a risk classification (HIGH RISK / LOW RISK) to the Safe Band MERN backend.
4. To integrate the risk classification output with the Safe Band wearable configuration system so that high-risk patients automatically receive enhanced monitoring parameters (increased ECG sampling frequency, prioritized nurse alerts, lower alarm thresholds).
5. To evaluate the prediction model against simulated Sri Lankan patient profiles, measuring precision, recall, F1 score, and false negative rate, with the objective of reducing false negatives compared to manual clinical judgment baselines.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) Methodology. The primary goal is to solve a real-world preventive healthcare problem by designing, building, and evaluating a functional artifact a Decision Tree based heart attack risk prediction model and its Flask API integration and measuring its performance against defined clinical benchmarks. This approach is chosen because the research contribution lies in the measurable improvement of pre-deployment patient risk stratification within the Safe Band IoT wearable system, evaluated through quantitative metrics including accuracy, AUC-ROC, and false negative rate.

3.2 Technical Methods and Algorithms

The prediction system will be developed using Python (scikit-learn) for the ML engine and Flask for API integration with the project's MERN stack backend.

- **Data Collection:** Patient risk factor data is collected via a structured Google Form capturing 13 fields covering demographics (age, gender, height, weight), lifestyle (smoking, physical activity), medical history (hereditary heart disease, diabetes, hypertension, current medication), self-reported health status, and cholesterol level. Responses are automatically exported from Google Sheets to the Python preprocessing pipeline.
- **Feature Engineering:** Raw responses are preprocessed into ML-ready features. BMI is computed from height and weight inputs ($BMI = \text{weight}(\text{kg}) / \text{height}(\text{m})^2$) and binned as Normal (0), Overweight (1), or Obese (2). Age is binned as above 50 (1) or 50 and under (0). All Yes/No responses are binary-encoded as 1/0; ordinal fields such as cholesterol level (Normal=0, Borderline=1, High=2) and health status (Good=0, Fair=1, Poor=2) are label-encoded.
- **Prediction Algorithm:** A scikit-learn DecisionTreeClassifier is trained using the Gini impurity criterion, maximum tree depth of 6, and minimum 10 samples per leaf to prevent overfitting. The tree is structured around clinically validated risk hierarchies: the root node splits on Hereditary Heart Disease, followed by Age, Smoking, Diabetes, Hypertension, Cholesterol, BMI, and Physical Activity at successive levels.
Output is a probability score and binary classification: HIGH RISK (YES) or LOW RISK (NO). Patients with probability between and flagged as BORDERLINE for manual nurse review.
- **Class Imbalance Handling:** SMOTE (Synthetic Minority Over-sampling Technique) from the Python imbalanced-learn library is applied to address the natural class

imbalance between high-risk and low-risk patients, ensuring the model does not under-predict high-risk cases during training.

- **Wearable Configuration Trigger:** The Flask API returns the risk label and probability score to the MERN backend. A HIGH RISK result automatically configures the Safe Band wearable into intensive monitoring mode activating increased ECG sampling frequency, lower alert thresholds, and prioritized display on the ward nurse dashboard.

3.3 Tools and Platforms

- **Machine Learning:** Python 3.x with scikit-learn for Decision Tree model training, hyperparameter tuning, 10 fold cross validation, and model serialization via joblib. Selected for its interpretability, lightweight deployment, and strong community support for healthcare ML applications.
- **API Layer:** Flask (Python) to expose the trained model as a REST API (POST /predict endpoint). Flask is selected for its simplicity, lightweight footprint, and seamless compatibility with the project's MERN stack backend integration via HTTP/JSON.
- **Data Collection:** Google Forms for structured patient questionnaire delivery, with Google Sheets API for automated response export. Chosen for its accessibility in Sri Lanka's clinical and community settings no special device or app required by patients or medical staff.
- **Backend Integration:** Node.js and Express.js (MERN stack) to receive prediction API responses and store risk profiles in MongoDB Atlas. Risk classification results trigger wearable configuration parameters via the project backend.
- **Development Environment:** Jupyter Notebook for exploratory data analysis, feature importance visualization, and model tuning. GitHub for version control of model code, preprocessing scripts, and Flask API. Google Cloud Platform (GCP) or AWS for cloud hosting of the prediction API during deployment.

3.4 Data Collection and Ethical Considerations

Model training data is sourced from the publicly available UCI Heart Disease Dataset (Cleveland subset, 303 anonymized records) supplemented by 500 synthetically generated Sri Lankan demographic profiles to improve regional representation. For live deployment, patient risk factor data is collected via the Google Form questionnaire. No personally identifiable information (PII) such as names or ID numbers is passed to the ML model at any stage; patient names collected in the form are used only for wearable assignment administration and are stripped before prediction inference.

All Google Form responses are stored in a SLIIT controlled Google Sheets account with restricted access. All API communications are encrypted via SSL/TLS protocols to maintain patient confidentiality, aligning with GDPR principles of data minimization, purpose

limitation, and security of processing. The prediction output is clearly communicated as a screening and wearable configuration tool only not a clinical diagnosis to both patients and medical staff.

3.5 Validation Strategy and Performance Metrics

- The system will be validated through a dual-evaluation strategy:
- Quantitative Evaluation: The Decision Tree model's accuracy, precision, recall, F1score, and AUC-ROC will be measured using 10-fold stratified cross-validation on the training dataset, then evaluated against 500 simulated Sri Lankan patient profiles.
Target benchmarks: Accuracy above, AUC-ROC above, Recall for HIGH RISK class (to minimize life-threatening false negatives), F1-Score above, and False Negative Rate below Results will be compared against a manual clinical judgment baseline to demonstrate measurable improvement in risk identification.
- Functional Testing: The Flask API's response latency (target: under 500 milliseconds per prediction), correct JSON input/output handling, and end-to-end integration with the MERN backend will be tested through simulated form submissions. Correctness of wearable configuration triggering HIGH RISK classification activating intensive monitoring mode will be verified against predefined test scenarios to ensure 100% correct configuration delivery for all high-risk patients.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution

The Safe Band project is a group research initiative developing an IoT-based wearable system for emergency health monitoring in Sri Lanka. The overall system comprises multiple components: real-time vital sign monitoring (Heart Rate, SpO2, ECG), an intelligent ward management dashboard, and the Prediction Component. The Prediction Component is the pre-deployment intelligence layer of the Safe Band ecosystem. Before a patient is assigned the Safe Band wearable device, their baseline cardiovascular risk is assessed using a structured questionnaire and a machine learning classifier. This ensures each patient's wearable is configured to the appropriate monitoring intensity from the very first use, shifting the entire system from purely reactive monitoring to a preventive-plusreactive approach.

The proposed solution consists of three tightly integrated sub-systems: a Google Formbased patient data collection pipeline, a Python Decision Tree prediction engine served via a Flask REST API, and a MERN stack integration layer that stores the risk classification and configures the Safe Band wearable accordingly. High-risk patients receive intensive ECG monitoring and prioritized alerts; low-risk patients are assigned standard monitoring parameters. This configuration-driven approach maximises the clinical value of the wearable hardware across the entire patient population.

4.2 System Diagram of Components

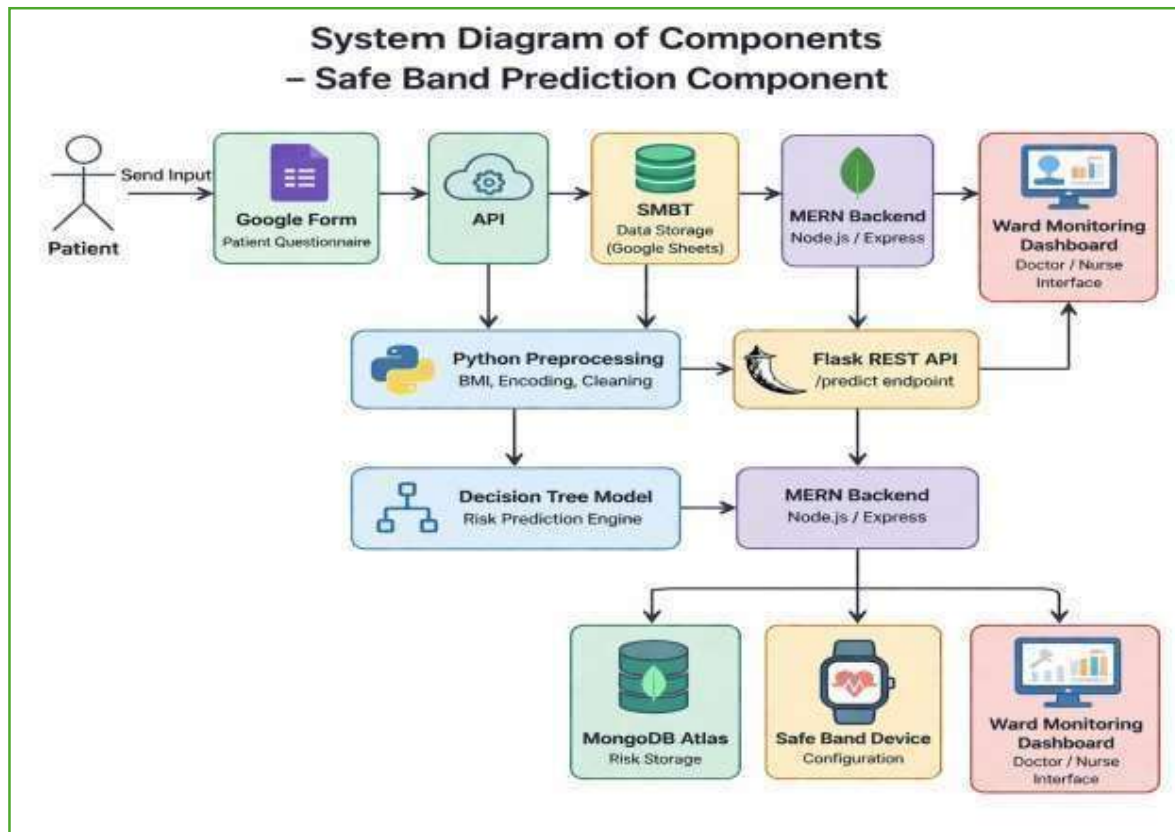


Figure 1: System Diagram of Components

4.3 Individual Component and its Sub-Parts

The individual focus of this research is the Heart Attack Risk Prediction Component. As shown in the system diagram, this component is built using Python (scikit-learn and Flask) and consists of three core sub-parts:

- **Data Collection and Preprocessing Module:** A structured Google Form captures 13 patient risk factor fields spanning demographics, lifestyle, medical history, and self-reported health status. Form responses are automatically exported via the Google Sheets API to a Python preprocessing script that computes BMI, applies age binning, and encodes all categorical inputs into ML-ready binary and ordinal features. This module is the entry point of the entire prediction pipeline.
- **ML Prediction Engine:** A Python-based scikit-learn DecisionTreeClassifier trained on the UCI Heart Disease Dataset (Cleveland subset) and supplemented with 500 synthetically generated Sri Lankan demographic profiles via SMOTE. The model is serialized using joblib and loaded at runtime by the Flask API. It accepts 11 preprocessed binary/ordinal features and outputs a probability score and a binary risk classification: HIGH RISK BORDERLINE, (flagged for nurse review), or LOW RISK.

- **Flask REST API Integration Layer:** A lightweight Flask application that exposes a POST /predict endpoint. It receives preprocessed patient feature data as JSON from the MERN backend, invokes the loaded Decision Tree model, and returns the risk classification result and probability score. Upon receiving a HIGH RISK classification, the MERN backend uses this result to configure the Safe Band wearable into intensive monitoring mode. This API layer bridges the Python ML world and the JavaScript/Node.js backend seamlessly.

4.4 Component Interactions

The system functions through a seamless interaction loop between the patient, the prediction pipeline, the MERN backend, and the wearable device. The following numbered steps describe each interaction in sequence:

- **Patient to Data Collection Module:** Before Safe Band device assignment, the patient completes the Google Form questionnaire. The form captures 13 risk factor fields and submits them to a secured Google Sheets spreadsheet via the Google Forms platform.
- **Data Collection Module to Preprocessing Script:** The Google Sheets API exports the new form response to the Python preprocessing script, which computes derived features (BMI binning, age binning, binary and ordinal encoding) and formats the data into a JSON feature vector ready for model inference.
- **MERN Backend to Flask API:** The MERN backend (Node.js / Express.js) sends the preprocessed patient feature JSON to the Flask REST API via an HTTP POST request to the /predict endpoint.
- **Flask API to Prediction Engine:** The Flask API passes the feature vector to the loaded Decision Tree model (joblib), which returns a probability score and risk classification (HIGH RISK / BORDERLINE / LOW RISK) within the 500-millisecond response target.
- **Flask API to MERN Backend:** The Flask API returns the JSON prediction response (risk label + probability) to the MERN backend. The backend stores the patient risk profile in MongoDB Atlas.
- **MERN Backend to Wearable Configuration:** Based on the stored risk classification, the MERN backend configures the Safe Band wearable device. HIGH RISK activates intensive monitoring mode (increased ECG sampling frequency, lower alert thresholds, priority display on nurse dashboard). LOW RISK applies standard monitoring parameters.
- **Backend to Ward Dashboard:** The risk classification and patient profile are pushed to the ward monitoring dashboard (developed by the ward monitoring component team

member). Medical staff can view each patient's predicted risk level alongside their real-time vitals, enabling informed clinical decisions.

4.5 Data Flow Explanation

The data flows through four distinct phases as it moves from patient input to wearable configuration output:

- **Ingestion Phase:** Raw patient data flows from the patient through the Google Form interface and is stored in Google Sheets. The Google Sheets API exports the response as structured JSON to the Python preprocessing script. At this stage, patient names are present in the data solely for administrative wearable assignment purposes.
- **Processing Phase:** The preprocessing script strips the patient name identifier, computes derived features (BMI binning, age binning), and encodes all categorical inputs into the 11-feature vector expected by the Decision Tree model. This anonymized feature vector is passed to the MERN backend, which forwards it to the Flask API via an HTTPS POST request. The Flask API invokes the serialized Decision Tree model and returns the prediction.
- **Storage and Configuration Phase:** The MERN backend receives the risk classification result (HIGH RISK / BORDERLINE / LOW RISK) and probability score. This result is stored in the MongoDB Atlas patient profile document and used to set wearable configuration parameters. BORDERLINE results are flagged for nurse review without automatic intensive configuration.
- **Visualization Phase:** The stored risk profile is pushed to the ward monitoring dashboard (React.js frontend) where the risk classification label is displayed alongside the patient's real-time vitals. Doctors and nurses can view the predicted risk level to contextualize any real-time alerts generated by the wearable device.

4.6 Scalability and Security Considerations

The Prediction Component is designed with both scalability and security as first-class concerns, ensuring it can grow from a research prototype to a production deployment across multiple hospital wards or community health programs.

- **Scalability:** The Flask API is stateless it holds no session state between prediction requests enabling horizontal scaling behind a cloud load balancer (GCP Cloud Run or AWS Elastic Load Balancer) to handle concurrent predictions during group or community-level deployment events. The MERN backend built on MongoDB Atlas natively supports elastic scaling of the patient risk profile database as the number of monitored patients grows. The Google Form data collection pipeline can accommodate any number of simultaneous patient submissions without infrastructure changes.

- **Data Security:** All communications between the Google Form pipeline, Flask API, and MERN backend are encrypted using HTTPS/SSL-TLS protocols to prevent interception of sensitive patient health data in transit. Google Sheets access is restricted to authorized project accounts only. Patient identifiers are stripped from the feature vector before any model inference occurs, ensuring the ML model never processes personally identifiable information.
- **Data Integrity:** MongoDB Atlas ensures data integrity for all stored patient risk profiles through built-in replication and journaling. The serialized Decision Tree model (joblib) is versioned in the GitHub repository so that any model update is traceable and reversible. Risk classification results stored in MongoDB are immutable audit records that can be reviewed for clinical quality assurance.
- **Access Control:** The Flask API is secured via API key authentication so that only the authorized MERN backend can invoke the prediction endpoint. The MERN backend enforces Role-Based Access Control (RBAC) with JWT authentication, ensuring that only authorized medical staff (Doctors, Nurses) and system administrators can view or act upon patient risk profiles stored in MongoDB.
- **Model Retraining and Maintainability:** As additional patient form data accumulates in the system over time, the Decision Tree model can be periodically retrained on the expanded dataset to improve accuracy and adapt to evolving Sri Lankan population health profiles. The Flask API is designed to hot-swap the serialized model file without downtime, enabling seamless model updates in production.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- **Patients:** Individuals completing the Google Form prior to Safe Band device assignment. They require a simple, non-technical questionnaire that takes under 5 minutes to complete.
- **Nurses and Medical Officers:** Clinical staff who receive the risk classification result on the Safe Band dashboard. They require accurate, interpretable risk labels with no false negatives for high-risk patients.
- **System Administrator (IT Staff):** Responsible for maintaining the Flask API and model versioning. They require a documented API schema and model retraining pipeline.

- Hospital Administrators: Interested in population-level risk statistics for ward planning and resource allocation.

5.2 Functional Requirements

1. FR1 : Questionnaire Data Collection: The system shall present a Google Form with 13 input fields covering demographics, lifestyle, medical history, and self-reported health status, and automatically export responses to a preprocessing pipeline.
2. FR2 : Feature Computation: The system shall automatically compute BMI from height and weight inputs, bin age into risk categories, and encode all categorical inputs into binary/ordinal features suitable for decision tree inference.
3. FR3 : Risk Prediction: The system shall classify each patient as HIGH RISK or LOW RISK based on the trained Decision Tree model, with a confidence probability score returned alongside the classification.
4. FR4 : Borderline Flagging: Patients with predicted probability between 0.4 and 0.6 shall be flagged as BORDERLINE and routed for manual nurse review rather than automated classification.
5. FR5 : API Integration: The Flask API shall expose a POST /predict endpoint that accepts patient feature JSON and returns the classification result within 500 milliseconds under normal load.
6. FR6 : Wearable Configuration Trigger: Upon receiving a HIGH RISK classification, the MERN backend shall automatically configure the Safe Band wearable to intensive monitoring mode (increased ECG sampling, lower alert thresholds).

5.3 Non-Functional Requirements

- NFR1 : Accuracy: The prediction model shall achieve a minimum accuracy and AUC-ROC on the validation dataset.
- NFR2 : Recall for HIGH RISK Class: The model shall achieve a recall of at least 90% for HIGH RISK patients to minimize life-threatening false negatives.
- NFR3 : Response Time: The Flask API shall return a prediction response within for a single patient submission.
- NFR4 : Interpretability: The decision tree model shall be exportable as a visual tree diagram, enabling medical staff to trace and audit any individual prediction path.
- NFR5 : Data Privacy: No personally identifiable patient names or identifiers shall be passed to or stored by the ML model inference pipeline.
- NFR6 : Availability: The Flask API shall maintain uptime during active deployment periods, supported by cloud hosting on Google Cloud Platform or AWS.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The immediate market is Sri Lankan hospitals, community health centers, and government health programs deploying IoT wearables for cardiac monitoring. The prediction component is directly relevant to any organization that issues wearable health devices and needs to prioritize monitoring resources based on patient risk profiles.

Customer Persona: Hospital Directors, Chief Medical Officers (CMOs), and Ministry of Health digital health program managers who are seeking cost-effective, technology-driven solutions to reduce preventable cardiac emergencies.

Secondary Market: Insurance companies offering health monitoring packages, elderly care homes, and occupational health programs for high-risk worker populations (e.g., construction, mining, or high-stress corporate environments).

6.2 Value Proposition

- For Patients: Ensures that wearable monitoring intensity matches their actual health risk, reducing the chance that a high-risk individual is monitored at an insufficient level.
- For Medical Staff: Provides an automated, evidence-based pre-screening tool that replaces time-consuming manual clinical assessments, freeing staff for higher-value tasks.
- For Hospitals: Reduces wasted monitoring resources on low-risk patients and reduces the likelihood of missed high-risk cases, lowering costs associated with preventable emergency admissions.
- For the Safe Band System: Adds a preventive intelligence layer that differentiates Safe Band from reactive-only wearable competitors.

6.3 Revenue Model

The Prediction Component will operate under a B2B Software as a Service model as part of the broader Safe Band platform:

- Per-Patient Assessment Fee: A micro-transaction model charging a small fee per risk assessment run through the platform.
- Annual Subscription: Hospital or health program subscription for unlimited assessments within a defined patient capacity tier.
- Integration License: A one-time fee for healthcare software vendors wishing to embed the prediction API into their own platforms.

6.4 Cost Estimation

- Development Costs: Python environment, scikit-learn model training, Flask API development covered by student research effort.
- Cloud Hosting: Flask API on Google Cloud Run or AWS Lambda estimated LKR 8,000 per month for initial deployment.
- Dataset Licensing: UCI dataset is open access; synthetic data generation is handled internally.
- Total Estimated Prototype Cost: LKR 25,000 to 35,000 (cloud hosting, testing tools, data simulation).

6.5 Pricing Strategy

Value-based pricing aligned with cost savings realized per prevented emergency admission. Tiered pricing: Basic (risk classification only), Standard (API integration + dashboard display), Premium (continuous model retraining + population analytics).

6.6 Competitive Advantage

- No Laboratory Tests Required: Unlike Framingham and similar tools, the Prediction Component relies solely on questionnaire data, making it accessible without specialist facilities.
- Interpretable Model: Decision trees allow clinical staff to trace and audit every prediction, building trust in AI-assisted screening.
- Integrated Wearable Configuration: Unique integration of risk prediction with IoT wearable configuration no commercially available competitor combines these two functions.
- Sri Lanka-Specific Calibration: Model tuned on Sri Lankan population profiles, addressing South Asian CVD phenotype differences.

6.7 Intellectual Property Considerations

- Copyright: Prediction model code, feature engineering pipeline, and Flask API will be protected under software copyright.
- Trade Secret: The specific decision tree feature weighting and calibration for Sri Lankan demographics will be maintained as a proprietary model configuration.
- Brand: The Safe Band brand and associated sub-components will be trademarked.

7. BUDGET AND JUSTIFICATION

Category	Item Description	Estimated Cost (LKR)	Justification
Software	Python environment, scikitlearn, Flask	0	Open-source; no licensing cost.
Data	UCI Heart Disease Dataset	0	Publicly available; open access.
Data Simulation	SMOTE synthetic data generation	0	Handled using Python imbalanced-learn library; no cost.
Cloud Hosting	Flask API on Google Cloud Run or AWS Lambda	12,000	Required to serve the prediction API to the MERN backend during testing and pilot deployment.
Testing Tools	500 simulated Sri Lankan patient profiles for evaluation	5,000	Data simulation and validation environment setup.
Google Forms	Google Workspace (if institutional plan not available)	8,000	Required for structured questionnaire deployment. SLIIT institutional account may cover this.
Human Effort	Model development, API coding, integration		Student individual research effort (no monetary cost assigned).
TOTAL		25,000	

Table 2 : Budget and justification

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS ID	Task / Sub-Task	Duration	Deliverable	Assigned Role
1.0	REQUIREMENT ANALYSIS AND RESEARCH	1 Month		
1.1	Identify the research problem: gap in predeployment heart attack risk screening for IoT wearables in Sri Lanka	1 Week	Problem Statement Document	Research Lead
1.2	Define cardiovascular risk factors from clinical literature and map to Google Form questionnaire fields	1 Week	Risk Factor Mapping Table	Research Lead
1.3	Identify functional and non-functional requirements for the prediction component and Flask API integration	1 Week	SRS Document	Research Lead
2.0	LITERATURE REVIEW	2 Months		
2.1	Review existing ML-based cardiovascular risk prediction models (Decision Tree, Random Forest, Naive Bayes)	2 Weeks	Annotated Bibliography	Research Lead
2.2	Study UCI Heart Disease Dataset structure, features, class distribution and suitability for Sri Lankan context	2 Weeks	Dataset Analysis Report	Research Lead

2.3	Study Design Science Research (DSR) methodology and evaluation frameworks for ML prediction components	2 Weeks	Methodology Chapter	Research Lead
2.4	Identify research gap: absence of questionnaire-driven pre-deployment risk classifier integrated with IoT wearables	2 Weeks	Research Gap Report	Research Lead
3.0	DATA COLLECTION AND PREPROCESSING	2 Months		
3.1	Design and deploy Google Form questionnaire (13 fields: demographics, lifestyle, medical history, health status)	1 Week	Google Form (Live Link)	ML Engineer

WBS ID	Task / Sub-Task	Duration	Deliverable	Assigned Role
3.2	Configure Google Sheets API to automate form response export to Python preprocessing pipeline	1 Week	Sheets API Integration Script	ML Engineer
3.3	Develop Python preprocessing script: BMI computation, age binning, binary encoding, ordinal encoding	2 Weeks	Preprocessing Script (Python)	ML Engineer
3.4	Apply SMOTE (Synthetic Minority Oversampling Technique) to generate 500 simulated Sri Lankan patient profiles	2 Weeks	Augmented Training Dataset	ML Engineer
4.0	MODEL DEVELOPMENT AND TRAINING	2 Months		
4.1	Train initial Decision Tree Classifier on UCI Heart Disease Dataset using Metrics	1 Week	Initial Model + Accuracy scikitlearn (Gini criterion)	ML Engineer

4.2	Perform hyperparameter tuning: max_depth, min_samples_leaf, criterion — using GridSearchCV	1 Week	Optimised Model Configuration	ML Engineer
4.3	Retrain model on SMOTE-augmented dataset combining UCI records and simulated Sri Lankan profiles	1 Week	Retrained Model File (.pkl)	ML Engineer
4.4	Validate model using 10-fold stratified cross-validation; measure accuracy, AUC-ROC, precision, recall, F1-score	2 Weeks	Cross-Validation Report	ML Engineer
4.5	Visualise Decision Tree structure and feature importance chart using sklearn.tree and matplotlib	1 Week	Decision Tree Diagram (Figure 2)	ML Engineer
5.0	FLASK REST API DEVELOPMENT	2 Months		
5.1	Set up Flask project structure, load serialised Decision Tree model (joblib), configure API key authentication	1 Week	Flask Project Environment	API Developer
5.2	Build and test POST /predict endpoint: accepts JSON features, returns risk label and probability score	2 Weeks	Working Flask API (v1.0)	API Developer

WBS ID	Task / Sub-Task	Duration	Deliverable	Assigned Role
5.3	Implement BORDERLINE case flagging logic (probability 0.4 to 0.6 routed for nurse review)	1 Week	Borderline Flag Module	API Developer
5.4	Configure SSL/TLS certificate and deploy Flask API to Google Cloud Run with HTTPS endpoint	1 Week	Live Cloud API Endpoint (HTTPS)	API Developer

6.0	INTEGRATION AND TESTING	1 Month		
6.1	Integrate Flask API with MERN backend (Node.js/Express.js): end-to-end form submission to risk classification			API Developer
6.2	Test wearable configuration trigger: verify HIGH RISK result activates intensive monitoring mode in Safe Band system	2 Weeks	Integrated System	API Developer
6.3	Unit testing and bug fixing: Flask API endpoints, preprocessing script, model inference edge cases	1 Week	Wearable Config Test Report	QA Engineer
		1 Week	Unit Test Report	
7.0	EVALUATION AND VALIDATION	1 Month		
7.1	Evaluate trained model on 500 simulated Sri Lankan patient profiles: precision, recall, F1-score, false negative rate	2 Weeks	Model Evaluation Report	Research Lead
7.2	Compare prediction accuracy against manual clinical risk assessment baseline; measure false negative reduction	1 Week	Comparative Analysis Document	Research Lead
7.3	Measure Flask API response latency under simulated concurrent load; verify sub- 500ms target	1 Week	Performance Metrics Report	QA Engineer
8.0	DOCUMENTATION AND FINAL SUBMISSION	1 Month		
8.1	Complete all dissertation chapters: Introduction, Literature Review, Methodology, Architecture, Results, Conclusion	3 Weeks	Final Thesis Document	Research Lead

WBS ID	Task / Sub-Task	Duration	Deliverable	Assigned Role
8.2	Prepare final presentation slides and viva demonstration of prediction component end-to-end pipeline	1 Week	Presentation Slides	Research Lead

Table 3: Work Breakdown Structure

9. GANTT CHART

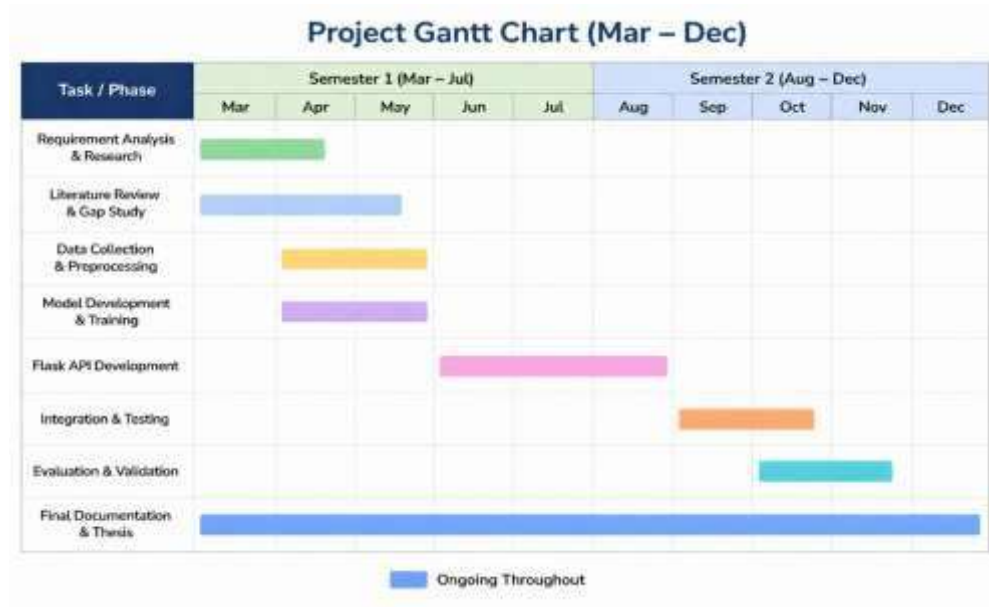


Figure 2: Gantt Chart

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11. APPENDICES

Appendix A: Survey Instrument Patient Risk Assessment Questionnaire

The following Google Form questionnaire is administered to every patient before Safe Band wearable device assignment. It collects the 13 risk factor fields that feed directly into the heart attack risk prediction model. The form is designed to be completed in under five minutes without requiring specialist medical knowledge or laboratory results. Patient names collected in Question 1 are used for administrative device assignment only and are stripped before model inference to preserve anonymity.

Section 1 : Demographic Information

- Full Name (Text field used for wearable assignment record only; not passed to ML model)
- Age in years (Numeric field used to compute age_bin: above 50 = High Risk)
- Gender (Multiple choice: Male / Female / Prefer not to say)
- Height in centimetres (Numeric field used with weight to compute BMI)
- Weight in kilograms (Numeric field used with height to compute BMI)

Section 2 : Lifestyle Information

- Do you currently smoke? (Multiple choice: Yes / No / Previously smoked but stopped)
- How many times per week do you exercise? (Multiple choice: 3 or more times per week / Less than 3 times per week / Rarely or never)

Section 3 : Medical History

- Do you have a family history of heart disease (hereditary)? (Multiple choice: Yes / No / Not sure)
- Have you been diagnosed with Diabetes (Type 1 or Type 2)? (Multiple choice: Yes / No)
- Do you have high blood pressure (Hypertension)? (Multiple choice: Yes / No / Not diagnosed but suspect I might)
- Are you currently taking medication for a heart or blood pressure condition? (Multiple choice: Yes / No)

Section 4 : Health Status and Cholesterol

- What is your current overall health status? (Multiple choice: Good / Fair / Poor)
- What is your most recent known cholesterol level? (Multiple choice: Normal / Borderline High / High / Not tested or unknown)

The questionnaire concludes with a data consent statement informing the patient that their responses will be used to configure their Safe Band wearable device monitoring parameters,

and that their data will be stored securely and used only for this research purpose in accordance with SLIIT ethical guidelines.

Appendix B: Detailed System Diagrams

This appendix contains the detailed logic diagrams for the Heart Attack Risk Prediction Component, illustrating both the end-to-end data flow pipeline and the decision tree classification structure.

Appendix C: Preliminary UI Mockups

The following mockups illustrate the proposed interaction design for the prediction component within the Safe Band patient intake workflow. These wireframes show how a nurse or medical officer would initiate the risk assessment, view the prediction result, and confirm the wearable configuration before device assignment.

Appendix D: Risk Analysis and Mitigation Table

The following table identifies the key technical, operational, data, and security risks associated with the development and deployment of the Prediction Component, together with the mitigation strategy for each risk.

Risk Category	Potential Risk	Impact	Mitigation Strategy
Technical	Decision Tree model overfitting on the small UCI Heart Disease Dataset (303 records), leading to poor generalisation	High	Apply regularisation parameters: max_depth=6, min_samples_leaf=10. Augment training data with 500 SMOTEGenerated Sri Lankan profiles. Validate using 10-fold stratified crossvalidation.

Risk Category	Potential Risk	Impact	Mitigation Strategy
Technical	on Sri Lankan patient profiles. detect		Monitor recall for HIGH RISK class to underfitting.
	Flask API response Google Cloud Run scaling enabled. asynchronous request testing with simulated submissions before community-level		Medium Deploy Flask API on latency exceeding the with auto-Implement 500-millisecond handling. Conduct target under load concurrent concurrent prediction pilot deployment. requests during deployment.
Technical	Google Sheets API rate limit reached during bulk form submission events (e.g., group hospital ward onboarding).	Medium	Implement request batching and exponential back-off in the preprocessing script. Cache recent API responses. Provide a CSV upload fallback pathway for bulk patient onboarding.
Operational	Patients providing inaccurate or incomplete questionnaire responses (e.g., unknown cholesterol level, uncertain	Medium	Set form fields for uncertain responses (e.g., "Not tested or unknown" for cholesterol). Route BORDERLINE cases (probability 0.4 to 0.6) to manual nurse review rather than automatic HIGH RISK classification. Provide clear field-level guidance text in the form.

Operational	hypertension diagnosis).		
Data	Resistance to adoption of AI-based risk screening among medical staff, reducing trust in prediction outputs.	Medium	Use an interpretable Decision Tree model (not a black-box) so clinicians can trace the exact reason path for any classification. Provide a "View Risk Factors" breakdown on the dashboard. Conduct staff orientation sessions demonstrating the tool as a screening aid, not a diagnostic replacement.
	Model providing inaccurate risk classifications (false	High	Apply SMOTE over-sampling to balance training classes. Optimise for HIGH RISK recall (target above 0.90) during
Risk Category	Potential Risk	Impact	Mitigation Strategy
negatives)	for hyperparameter highrisk patients imbalance in over training data.	tuning. due the time.	Regularly retrain model as new patient form data to class accumulates to improve performance
Data	UCI Heart Disease Dataset features may not fully represent the South Asian cardiovascular risk phenotype, reducing accuracy for Sri Lankan patients.	High	Supplement UCI training data with 500 synthetically generated Sri Lankan demographic profiles using SMOTE. Evaluate the final model specifically on the simulated Sri Lankan profiles. Plan for real patient data collection in Semester 2 to progressively replace synthetic profiles.
Security	Unauthorised access to patient risk profiles stored in MongoDB Atlas.	High	Enforce Role-Based Access Control (RBAC) and JWT authentication on all MERN backend API routes. Restrict MongoDB Atlas cluster access to whitelisted IP addresses. Conduct security audit before pilot deployment.

Security	Interception of patient health data in transit between the Google Form pipeline, Flask API, and MERN backend.	High	Encrypt all communications using HTTPS/SSL-TLS protocols. Use API key authentication on the Flask /predict endpoint so only the authorised MERN backend can invoke it. Strip patient identifiers from feature vectors before model inference.
Ethical	Patients misinterpreting the HIGH RISK classification as a confirmed medical diagnosis, causing unnecessary anxiety.	Medium	Include a clear disclaimer in both the Google Form and the dashboard result display: "This result is a screening and wearable configuration tool only — it is not a medical diagnosis." Route BORDERLINE results for nurse review. Train medical staff to communicate the tool's purpose and limitations clearly to patients.

Table 4: Risk Analysis and Mitigation

Appendix E: Ethical Compliance Statement

As this research involves the collection and processing of sensitive patient health data, the following ethical guidelines are strictly observed throughout the design, development, and deployment of the Heart Attack Risk Prediction Component.

Data Minimisation and Purpose Limitation

- Only the 13 risk factor fields strictly necessary for cardiovascular risk prediction are collected via the Google Form. No unnecessary personal data (national ID, address, financial information) is requested at any stage.
- Patient names are collected in the form solely for the purpose of wearable device administrative assignment. Names are stripped from the data pipeline before any ML model inference occurs and are never stored in the prediction result records.
- All collected data is used exclusively for the purpose of configuring Safe Band wearable monitoring parameters and for academic research evaluation within the SLIIT research project.

Anonymisation and Data Security

- All training and validation datasets used for model development are fully deidentified. The UCI Heart Disease Dataset is an anonymised public dataset. The 500 synthetically generated Sri Lankan profiles contain no data corresponding to real individuals.
- All API communications between the Google Form pipeline, Flask API, and MERN backend are encrypted using HTTPS/SSL-TLS protocols to prevent interception of patient health data in transit.
- Patient risk profiles stored in MongoDB Atlas are accessible only to authorised medical staff (Doctors and Nurses) via Role-Based Access Control (RBAC) with JWT authentication. No patient data is shared with third parties.

Transparency and Informed Consent

- The Google Form concludes with an informed consent statement clearly explaining to the patient: (1) the purpose of the data collection, (2) how the data will be used to configure their wearable device, (3) that participation is voluntary, and (4) that the classification result is a screening and prioritisation tool, not a clinical diagnosis.
- The prediction output is clearly labelled on the nurse dashboard as a screening result. BORDERLINE cases are routed to nurse review rather than automatic device configuration. Medical staff are trained to communicate the tool's purpose and limitations to patients.

Regulatory Compliance

- The system architecture is designed to align with the key principles of the General Data Protection Regulation (GDPR): lawfulness and fairness, purpose limitation, data minimisation, accuracy, storage limitation, integrity and confidentiality, and accountability.
- No personally identifiable information (PII) is processed by the machine learning model at any stage. All model inference operates exclusively on anonymised binary and ordinal feature vectors.
- All research activities involving patient data collection are conducted in accordance with SLIIT institutional ethical guidelines and the applicable data protection regulations of Sri Lanka.
- System performance metrics (accuracy, recall, F1-score, false negative rate) are recorded and reported with full transparency in the research evaluation. No results are selectively reported or omitted.

